L1.3 Continuity

Work in your group. Check your answers against the Quarto tonight.

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1 · What continuity is

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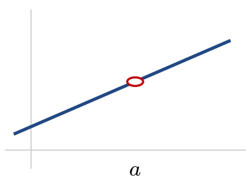
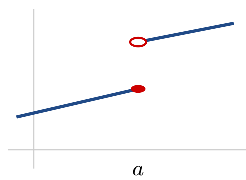
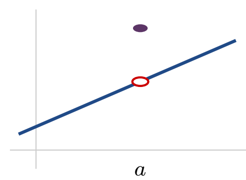
1. For each graph, say which of the three parts of LHL $= f(a) =$ RHL hold at $x = a$, and which fail. **Answer in words** — *the limit*, *the value*, *the match*.



(a)



(b)



(c)

2. If f and g are both continuous at $x = a$, is $h(x) = f(x) + g(x)$? Fill the three blanks:

$$\lim_{x \to a} h(x) = \lim_{x \to a} \big(f(x) + g(x)\big) = \underline{\hspace{6cm}} = \underline{\hspace{5cm}} = \underline{\hspace{3cm}}$$

One of those steps is a **limit law** you already had. One of them is **today's definition**, used as a substitution. Say which is which.

2 · Where continuity breaks

3. $f(x) = \dfrac{x^2-1}{x^2+x-2}$

(a) Factor, then give `HE @`, `Z @`, `VA @` and `HA @`.

(b) State the domain D — both notations.

(c) **Now say where f is continuous.** How much work did (c) take, given (b)?

4. For each function: is it continuous at the marked point? If not, name the kind and say **what died** — the limit, the value, or the match. **One of these four is continuous.**

(a) $\dfrac{x^2-2x-3}{x-3}$ at $x=3$ (b) $|2x-1|$ at $x=\frac{1}{2}$ (c) $\dfrac{1}{x-5}$ at $x=5$ (d) $\lfloor x \rfloor$ at $x=2$

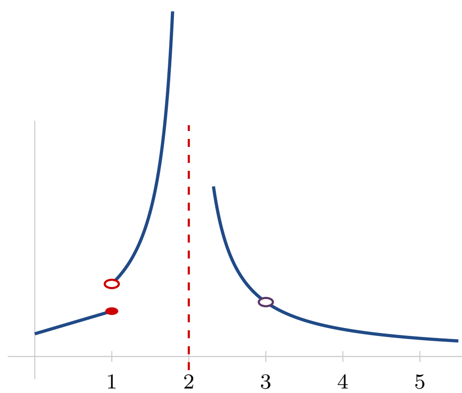
> **Stop.** Before you classify anything — write down the one equation from memory.

5. Is $\sqrt{x-1}$ continuous on $[1,2]$?

(a) What goes wrong if you try to run the two-sided test at $x = 1$?

(b) What has to change, and why is that not cheating?

6. On which intervals is f continuous?



Then: **one of your brackets is square and the rest are round. Which, and why?**

7. Are there values c and m that make f continuous at $x = 1$? Find them, or explain why they cannot exist.

$$f(x) = \begin{cases} cx^2 & x < 1 \\ 4 & x = 1 \\ -x^3 + mx & x > 1 \end{cases}$$

3 · Intermediate Value Theorem

8. Show that $f(x) = 4x^3 - 6x^2 + 3x - 2$ has a root in $[1, 2]$.

Do not just give the answer — **state the hypothesis you checked**, and say what the theorem gives you and what it does not.

9. Show that $x^5 = 1 - x$ has a solution in $[0, 1]$.

> **Stop.** The theorem only ever finds a *root*. So what function do you need to build first?

10. BOARDS A number whose **cube is one more than itself**, somewhere on $[0, 2]$.